Do Now · retrieval

Recursion: call stack and tracing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Factorial(n) returns 1 when n=0; otherwise it returns n*Factorial(n-1). Trace the successive argument values in Factorial(3), stopping when the base case is reached.

Q2

What is the difference between a function calling another function and returning a value to its caller?

Q3

A stack contains bottom to top P, Q, R. After one pop, what is at the top?

Q4

A function Square(x) returns x*x. Find Square(2)+Square(3), showing both calls.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Build the stack — SumTo(4) waits for SumTo(3), which waits for SumTo(2), which waits for SumTo(1).

Core idea 2

Separate the local values — Each stack frame holds its own n: 4, 3, 2 or 1. The values do not overwrite one another.

Core idea 3

Unwind from the top — SumTo(1) returns 1; SumTo(2) returns 3; SumTo(3) returns 6; SumTo(4) returns 10.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

stack frame — An entry containing the state for one active function call.

unwind — Return from calls in reverse order after a base case is reached.

local variable — A variable owned by a particular invocation.

call stack — Last-in, first-out storage for active calls.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

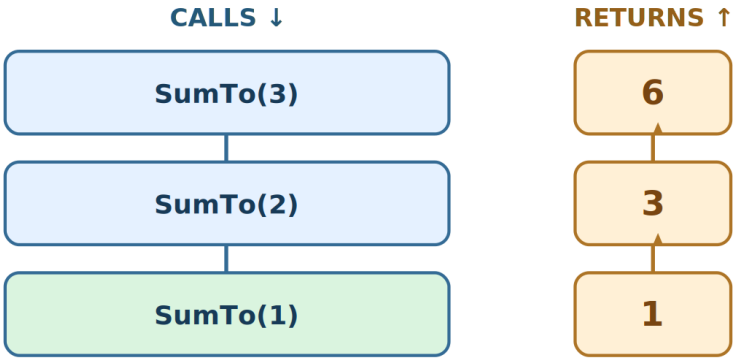
Problem

Draw the call-stack frames for SumTo(3). Which call returns first?

Follow the worked method

1. Push — The frames are SumTo(3), SumTo(2), then SumTo(1), with SumTo(1) at the top.
2. Stop — SumTo(1) is the base case; it returns 1 first.
3. Unwind — SumTo(2)=2+1=3, followed by SumTo(3)=3+3=6.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Call sequence
SumTo(5) → SumTo(4) → SumTo(3) → SumTo(2) → SumTo(1).

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

List the call-stack frames from bottom to top for SumTo(4).

Question 2

Trace the returned values for SumTo(6).

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A pupil writes that SumTo(4) returns 4+3+2+1 before SumTo(1) has executed. Explain the order error.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Every recursive call uses the same local-variable storage.

Use key vocabulary from memory

1. Explain stack frame in your own words.

2. Explain unwind in your own words.

3. Explain local variable in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.